Unit II: Plant Morphology and Taxonomy of Angiosperm

Chapter 4

Reproductive Morphology

Learning Objectives

The learner will be able to,

- *List the types of Inflorescence.*
- *Distinguish Racemose and Cymose inflorescence*
- *Dissect a flower and explore the parts of a flower.*
- *Compare various types of Aestivation.*
- *Explore various types of Placentation.*
- *Understands the types of Fruits and seeds*
- *Learn the structure of fruits and seeds*
- *To differentiate Monocot and Dicot seeds*

Chapter Outline

4.1. Inflorescence
4.2. Flower
4.3. Accessory organs
4.4. Androecium
4.5. Gynoecium
4.6. Construction of floral diagram and floral formula
4.7. Fruits
4.8. Seed

Flowers have been a universal cultural object for millennia. They are an important aesthetic element in everyday life, and have played a highly symbolic role in our culture throughout the ages. Exchange of flowers marks respect, affection, happiness, and love. However, the biological purpose of the flower is very different from the way we use and perceive. Flower helps a plant to reproduce its own kind. This chapter discusses flowers, their arrangement, fruits and seeds which are the reproductive units of a plant.

Floriculture

Floriculture is a branch of Horticulture. It deals with the cultivation of flowers and ornamental crops. The Government of India has identified floriculture as a sunrise industry and accorded the status of 100% export oriented. Agriculture and Processed Food Product Export Development Authority (APEDA) is responsible for export promotion of agricultural and horticultural products from India.



4.1 Inflorescence

Have you seen a bouquet being used during functions? Group of flowers arranged together on our preference is a bouquet. But an inflorescence is a group of flowers arising from a branched or unbranched axis with a definite pattern. Function of inflorescence is to display the flowers for effective pollination and facilitate seed dispersal. The grouping of flowers in one place gives a better attraction to the visiting pollinators and maximize the energy of the plant.

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